How Gaussian Data Can Produce Heavy-Tailed SGD Iterates

Mert Gürbüzbalaban, Umut Şimşekli and Lingjiong Zhu (ICML 2021)

The Heavy-Tail Phenomenon in SGD

Key: gurbuzbalaban2021heavy Tags: D / S; B Tier: Must read

Survey: Dynamics

Spreadsheet priority: None; independent recommendation: Must read

The paper in one minute

Question: Can SGD develop power-law fluctuations even when the input data have light tails?

Model/Method: View constant-step streaming least-squares SGD as a random affine recursion, with random curvature multiplying the current iterate.

Finding: Under contraction and nondegeneracy conditions, the stationary law has a power-law tail; its variance can be infinite despite finite moments of every order in the data.

Why we care: Noise is state dependent. Learning rate, batch size, dimension and curvature affect long-run fluctuations; this is not itself an efficient saddle-escape or generalization theorem.

1. Separate the data, the noise and the iterates

A light-tailed dataset does not force light-tailed optimizer iterates. The reason is multiplicative randomness: the sampled Hessian rescales the current parameter at every step. Repeated products of mostly shrinking, occasionally expanding factors can produce large excursions. An additive Gaussian approximation with a fixed deterministic contraction misses this mechanism.

$$F(x) = (1/2)E[(a^T x - y)^2],$$

$$x_k = M_k x_{k-1} + q_k,\ M_k = I - (\eta/b)\Sigma_i\, a_i a_i^T,\ q_k = (\eta/b)\Sigma_i a_i y_i.$$

Each new batch contains fresh, independent data. Thus the pairs (M_k, q_k) are independent across k and independent of the previous iterate. M_k and q_k within a batch need not be independent. This streaming assumption is central: a fixed finite dataset revisited for many epochs is a separate empirical setting.

2. What a tail index means

A power-law tail $P(|X| > t)$ behaving like $c\, t^{-\alpha}$ assigns much greater probability to extreme values than a Gaussian. Smaller α means heavier tails. Integrating the tail shows why a pth moment is finite below α and diverges at or above α for a genuine nonzero power-law tail. In particular, $1 < \alpha < 2$ permits a mean but no finite variance. A power-law stationary law need not itself be an exactly α-stable distribution.

Convergence of laws, not parameter convergence. The distribution can approach a stationary law while individual constant-step iterates keep fluctuating. A stationary random parameter is not an optimizer that has settled at a single point.

3. Worked example: stable distribution, infinite variance

Pedagogical illustration, constructed for these notes.

Take independent a,y ~ N(0,1), scalar x, batch size b = 1 and η = 1. The population objective is $F(x) = E[(ax-y)^2]/2 = (x^2+1)/2$, a strictly convex quadratic with minimizer zero. The stochastic update is $x^+ = (1-a^2)x + ay$.

$$\textit{Write } M = 1-a^2\textit{. Its second-moment multiplier is } E[M^2] = 1 - 2E[a^2] + E[a^4] = 1 - 2 + 3 = 2.$$

A multiplier larger than one does not alone rule out a stationary distribution. Contraction of random products is controlled by E log|M|. Let φ(z) be the standard Gaussian density. Since E[M] = 0, symmetry and integration by parts give the following exact first absolute moment.

$$E|1-a^2| = 4\int_0^1 (1-z^2)\varphi(z)dz = 4[z\varphi(z)]_0^1 = 4\varphi(1) \approx 0.967883 < 1.$$

Jensen gives E log|M| ≤ log E|M| < 0, so the multiplicative dynamics contract on average. In the scalar tail equation $h(s) = E|M|^s$, we have h(1) < 1 and h(2) = 2 > 1. The nonzero root h(α) = 1 therefore lies strictly between 1 and 2. The continuous Gaussian setup supplies the needed nondegeneracy and moment conditions.

Consequently the stationary SGD law has a finite mean and infinite variance, although both a and y have moments of every order. For finite k and $x_0 = 0$, independence and E[y] = 0 give $E[x_k^2] = 2E[x_{k-1}^2] + 1 = 2^k - 1$. Finite-time moments exist while growing without bound; there is no contradiction with convergence in distribution.

The scalar variance threshold in Proposition 5 is $\eta_{crit} = 2b/[\sigma^2(d+b+1)] = 2/3$. Our η = 1 lies above it, but stationarity is established separately by the absolute-moment calculation. A large step should not be assumed stable merely because it exceeds the variance threshold.

4. The stationary-law theorem

For matrix products $\Pi_k = M_k \cdots M_1$, define the top Lyapunov exponent as their asymptotic logarithmic growth rate and h(s) as $\lim(E\|\Pi_k\|^s)^{1/k}$. Theorem 2 states a unique stationary law and directional power-law tails when the exponent is negative and a unique positive α solves h(α) = 1, under the paper's continuous-data and nondegeneracy setting. These conditions distinguish average contraction from rare expanding products.

With isotropic Gaussian $a \sim N(0,\sigma^2 I)$, Theorem 3 simplifies the product quantities to $E \log\|Me_1\|$ and $E\|Me_1\|^s$. Rotational symmetry makes the distribution of $\|Mv\|/\|v\|$ independent of the direction of v. This is a fixed-direction moment. Do not replace it by $E\|M\|^s$, the one-step operator-norm moment, which is a different quantity and only supplies a general bound.

$$h(2) = 1 - 2\eta\sigma^2 + (\eta^2\sigma^4/b)(d+b+1).$$

Solving h(2) = 1 gives $\eta_{crit} = 2b/[\sigma^2(d+b+1)]$. Below it, the Gaussian setting has α > 2. At the threshold, α = 2. Just above it, while average contraction still holds, α < 2. Beyond stability, convergence is no longer guaranteed. Theorem 4 gives increasing α with batch size and decreasing α with step size or input variance when α ≥ 1, plus decreasing α with dimension. It does not say η/b alone completely determines the distribution.

5. Rates and averaging have moment restrictions

Theorem 8 bounds the p-Wasserstein distance between the current and stationary laws by $h(p)^{k/p}$ times the initial distance, for $1 \le p < \alpha$. Wasserstein distance compares distributions by the cost of transporting one to another; W_2 is inappropriate when the stationary variance is infinite. Corollary 11 describes stable-law limits of appropriately centered sums when α < 2, with a special normalization at α = 2. It does not make every finite average exactly stable or Gaussian.

6. What the experiments measure

Synthetic experiments revisit fixed Gaussian datasets and estimate tail indices of centered, averaged runs. Figure 1 varies η, b, dimension and input variance; larger steps and smaller batches generally yield heavier estimated tails. RMSProp and SVRG comparisons concern the tested implementations and do not constitute a general theorem about those optimizers.

Figures 2-3 test fully connected ReLU networks on MNIST/CIFAR10 and an 11-layer VGG network on CIFAR10. Runs use 10,000 iterations and average the last 1,000. Because repeating whole-network experiments thousands of times is expensive, layer coordinates are treated as an approximately independent collection for tail estimation. This is a modeling approximation, not a proved finite-network independence result. Some layers show a stronger η/b trend than others; VGG estimates above two are clipped to two in the plot.

7. What remains outside the theorem

The sharp theory is for constant-step streaming linear regression. Deep-network evidence is empirical; repeated finite-data sampling, changing learning rates and adaptive optimizers require further analysis. The paper discusses connections to flatness and generalization, but does not prove that choosing heavier tails always improves test error. It explicitly leaves the effect of tail index on saddle-escape time as future work. Several printed auxiliary formulas contain slips; these notes rely on the stated fixed-direction identity, relevant theorem qualifications and the independently derived scalar example, rather than certifying every supporting inequality.

8. Survey connection and reference table

Fang2019 controls finite-time escape using bounded, dispersive stochastic noise. This paper studies a stationary law generated by random curvature, where global bounded-noise assumptions may fail. Azizian2025 studies rare transitions and expected global hitting times under a different controlled-noise model. A stationary heavy tail cannot simply be inserted into either theorem without checking assumptions.

Quantity	Interpretation
$E \log\|\|\Pi_k\|\| / k$	Typical product growth; negative supports stationarity.
$h(\alpha) = 1$	Boundary between controlled and divergent moments.
$1 < \alpha < 2$	Stationary mean exists; variance diverges.
η_{crit}	Variance boundary, not the upper stability limit.

Source: publisher main PDF1-9, Theorems2-4/8, Proposition5, Corollary11; supplement1-16, especially C.1-C.6 and D. Scalar Gaussian calculation constructed for teaching.

Papers Master - Our notes (copy): Constant-step streaming least-squares SGD can have a power-law stationary law, including infinite variance despite light-tailed inputs. Random curvature supplies multiplicative noise. Gaussian-input tail-index identities and monotonicity have specific assumptions; finite-data neural-network evidence and generalization interpretations remain separate.

Reading recommendation: Read the affine-recursion mechanism and distinguish streaming quadratic theorems from finite-data neural-network measurements. This adds a different perspective on SGD noise.